Spiny lobsters fisheries in the Western Central Atlantic

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by

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Abstract: Spiny lobsters annual average catches in 35 fishing areas during the 1936-2015 period are positively correlated (P < 0.0001) with their surfaces (km^2). Recent landings, when expressed as peak catches percentages, exhibit a negative significant correlation (P < 0.006) with the number of years after peak landings, and have decreased on average 38%, reflecting abundances and efforts reductions of 48% and 44 %, respectively. Annual average yields (kg/km^2) are significantly correlated with fishing areas surfaces at different levels (P < 0.03 to P < 0.004) depending on the fisheries' evolution moment. Annual average yields in theoretically considered larval importing areas were not different (P > 0.11) to those in larval exporting areas, but were greater (P < 0.002) in "downstream" (Western) than in "upstream" (Eastern) Caribbean areas underscoring the prevalent role geography plays in these fisheries' yields. Fifteen producers, accounting for 97% of 1950-2013 landings, exported 68% of their combined catches during 1985-2015. That *P. laevicauda* has a westward declining abundance from Brazil to Barbados, Venezuela, Cuba, Florida, and Bermuda suggests a larval influx via the Guiana/South Equatorial currents as they flow along the Brazilian coast and north into the Caribbean. Pre-recruit (juvenile) generalized harvest and commercialization represent about 25% of all *P. argus* landings and, along with inconsistent regulations' implementation, are the two major regional problems. With high demand and prices, this is the right moment for all spiny lobster fisheries to regulate fishing effort and develop adequate protection measures agreed to and enforced on a regional basis, because they all are either "fully exploited" or "overfished".

Key words: peak landings; fisheries yield; effort reduction; postlarval influx; *Panulirus* fisheries.

Introduction

Background

According to the FAO (2014) world fisheries catch summaries, the "Caribbean spiny lobster" (*Panulirus argus*) accounted for 74% of the total 2.42 million metric tons (M mt) landings of all *Panulirus* species during the 1950-2013 period, followed by the Australian "Western rock lobster" (*P. cygnus*) with 25%, and the remaining 3% shared by 19 other congeneric species' harvests. In the Caribbean Basin *P. argus* fisheries represent 80% (1.44 M mt) of the species' world total with the remaining 20% (0.35M mt) harvested in Brazil, and presently

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generates about $500 million (USD) combined annual income by an estimated 100,000 commercial fishermen, with a total annual export value of close to $900 million (USD) for the participating countries (FAO, 2015).

Fisheries directed at *Panulirus argus* and two other congeneric species (*P. guttatus* and *P. laevicauda*) throughout their geographical range, have declined from over 52,000 mt maximum combined annual harvest, to presently about half, with individual fisheries' most recent landings being 2% to 93% of their individual maxima, depending on the number of years after perking.

The long time held "logical hypothesis" about the possibility of *P. argus* larval connectivity between all Caribbean spiny lobster populations due to its lengthy planktonic period (Lewis, 1951) has now been confirmed for all studied *P. argus* populations are both genetically mixed and impossible to individualize (Tringali et al., 2008); larvae connectivity probabilistic studies add theoretical indirect support to the genetic results (Cowen et al., 2002; Kough et al., 2013).

Presently most Caribbean countries enforce some type of regulation with different degrees of consistency in an attempt to individually protect their spiny lobster stocks which, regionally, are considered to be either "fully exploited" or "overfished" (FAO, 2001). A major challenge to resource managers is that their regional larval interconnectivity impede accurate local parental based evaluations; leaving annual recruit classes' sizes calculations based on postlarval arrival and survival rate, along with biological, and fishing effort regional regulations, as the only viable basis for management alternatives to enhance their long-term survival.

The significance of this cumulative evidence from genetic and connectivity studies resulted in the "St. George's Declaration" signed at the 9th meeting of the Caribbean Regional Fisheries Mechanism (http://www.fao.org/ fi/ staticmedia/MeetingDocuments/WECAFC16/8e.pdf), in May 2015 (Barbados). The "St. George's Declaration" is aimed at collectively managing *P. argus* fisheries through the Caribbean, proposed to be based *"...on the best scientific evidence available"* which, although non-binding is another step in the right direction, and adds-up to other regional efforts already in place such as the collective March 1st to June 30th closed season that Panama, Costa Rica, Nicaragua, Honduras, Guatemala, Belize, and the Dominican Republic observed in 2017 for the 8th consecutive year (http://www.clmeproject.org/2017/03/02/regional-closed-season-for-the-caribbean-spiny lobster/).

Objectives

Special emphasis has been given to the harvesting areas rather than to the reporting countries totals because some areas had been harvested simultaneously by several countries. Compilations of landings by harvest area resulted in calculating new totals, different to those reported by FAO (2014; 2015), for Bahamas, Belize, Bermuda, Brazil, Colombia, Cuba, the Dominican Republic, Florida, Honduras, and Nicaragua, and separating Jamaica's proper and Pedro Bank's harvests. The objectives of this report are to analyze and discuss the evolution of each Caribbean and the Brazilian spiny lobster fisheries' current status and perspectives, the role of fishing effort in Bermuda's *P. argus* and *P. guttatus* fisheries recovery, in Brazil's *P. laevicauda* fishery collapse, and in all *P. argus* populations' situation, along with suggestions about their management.

Material and methods

Caribbean 1950-2013 landings can be found at http://www.fao.org/fishery/statistics/global-commodities-production/3/en (FAO, 2014); and 2014 landings in FAO (2015). Florida and California landings (1950-2015) are at the NOAA (National Oceanic and Atmospheric Administration) web-site (https://www.st.nmfs.noaa.gov/commercial-fisheries/commercial-landings/annual landings /index); Florida's total landings were reduced by 22,700 tons harvested in international waters during 1964-1979 (Labisky et al., 1980). Fisheries Statistics for the Western Pacific (Hawaii, American Samoa, Commonwealth of Northern Marianas Islands, and Guam) for 1985-2015 are at the NOAA (https://www.pifsc.noaa.gov/wpacfin/reportlanding.php).

Insular and continental shelves surface areas (km^2), as well as some fisheries characteristics and other data, are from FAO countries' profiles (www.fao.org/fi/website/FISearchAction.do, CRFM, 2011; FAO 2001; 2003; 2007; 2011; 2015), and from some regional reports (Munro, 1983; Olsen et al., 1984). Although "culturally" and traditionally Bermuda is considered part of the Caribbean Basin at large, its fishery is analyzed separate from those within the Caribbean geographical boundaries; some major Caribbean producers', and Brazilian fishery are described individually also.

Microsoft Excel 2003 was used to calculate descriptive statistical parameters, and figures; the Vassar College computation website (http://www.vassarstats.net/vst.html) was accessed to complete all significance tests and correlations, always using log transformed data. The correlation between most recent landings expressed as maximum harvests percentages to years from peaking, was calculated using every year class after peaking percent averages (435 individual values). Verbatim quotations are presented *in italics* throughout the text.

Results

Included species

In the Western North Atlantic three congeneric *Panulirus* spiny lobsters, with different but to some overlapping geographical ranges, are recreationally or commercially harvested at different levels, along with a fourth and more scarce spiny lobster sporadically present in Brazilian catches.

Panulirus argus (Latreille, 1804), designated as "Caribbean spiny lobster" in FAO landings reports, is found from Bermuda (32.53º N), all along the Americas' Eastern continental coasts, from Beaufort, North Carolina (34.71° N), the whole Gulf of Mexico, and every Caribbean continental coast and island, to Río de Janeiro, Brazil (22.90° S), being the only *Panulirus* reported by Nizinski (2003) in her crustaceans from the Atlantic coast of the United States list. The species has also been reported twice from Ivory Coast, West Africa (4° - 9° W) (Holthuis, 1991), along with a permanent population in the Central-Eastern Atlantic Cabo Verde Archipelago (22.90° - 24.92° W) commercially exploited for the past 15 to 20 years (Freitas & Castro, 2005). The Caribbean spiny lobster has the greatest level of landings among all spiny lobsters' worldwide as a consequence of its large geographical range, the widest among all commercially exploited spiny lobsters. During the 1950-2013 period its total landings reached 1.79 M tons, with 80% harvested in the Caribbean Basin, and the remaining 20% in Brazil.

Panulirus guttatus (Latreille, 1804), named "spotted spiny lobster" by FAO, is essentially a Caribbean Basin limited spiny lobster, found from Bermuda (32.53 ° N), where is commercially harvested, to Bahamas, South Florida, Belize to Panamá, and the Caribbean arc from Cuba to Trinidad, Curaçao, Bonaire, Los Roques, and Suriname (5.15 ° N) (Holthuis, 1991). Bermuda's *P. guttatus* landings during 1976-2015 were 179 tons or 14% of its spiny lobsters total landings. Martinique reported 1.3 tons representing 10% of its total spiny lobsters combined landings during the 2009-2013 period and, if the report that 5% of Anguilla's spiny lobster harvest are "guinea chick" is accurate (Wynne and Côté, 2007), 243 additional tons could be added during the 1974-2013 period to this species harvest, for a Caribbean total of about 450 tons, which most likely is a regional underestimate for most Caribbean islands catching this species, either directly or as by-catch (Honduras-Nicaragua, Jamaica, the Dominican Republic, Barbados' and Dominique's east coasts, Montserrat, and Trinidad) have incomplete fisheries statistical information.

Panulirus laevicauda (Latreille, 1817), designated "smooth tail spiny lobster" by FAO, has been reported from Bermuda and Florida, Yucatán and most Caribbean islands, down to Eastern Brazil (16 ° S) (Holthuis, 1991) with 41,440 tons total catch, equivalent to 11% of Brazil's total spiny lobsters landings during the 1970-2013 period (FAO, 2014), which most certainly is a large underestimate for it has always been harvested along with *P. argus,* and during the 1961-1969 period constituted 40% of their combined landings (Buesa and Paiva, 1969) which could represent at least 13,000 additional tons. Unquantified harvests are also reported from Colombia (La Guajira), Costa Rica, the Dominican Republic, Saint Lucia, and this species is considered to be Barbados' second most abundant spiny lobster (after *P. argus*).

A fourth spiny lobster, *Panulirus echinatus* Smith, 1869, named "brown spiny lobster" by FAO, is known from the Central Atlantic Islands, to its 1869 type locality in Pernambuco State, Brazil (Holthuis, 1991), occasionally appearing in the Recife area catches (Góes et al., 2007).

Bermuda fisheries

Commercial landings reports

Bermuda's *P. argus* population, at the species' outermost northern geographical range limit, excluded from existing theoretical connectivity calculations, and most likely solely dependent on Caribbean imported post-larvae for its survival, sustains an old, stable, and very well managed fishery (Trott et al., 2002). According to FAO (2014) landings data, during the 1950-2013 period *P. argus* harvests totaled 5540 mt, 83% of which correspond the 1950-1978 period, including a 250 mt maximum in 1968 followed by 159 mt annual average landings, but it is imperative to note that FAO landings for 1950-1973, are deemed excessive, of unknown source (Luckhurst et al., 2003) and inconsistent with the period's fishing effort and market capabilities [(1)]. Further, the 1946 Colonial Annual Report for Bermuda (Anonymous, 1948) includes a catch equivalent to 35 mt while, according to FAO (2014), just 4 years later, the catch was 4 times larger (150 mt), which is highly improbable.

(1) J.M. Pitt - Bermuda Dpt. of Environment and Natural Resources. Personal communications, 2017

Consequently the 4400 mt landings reported by FAO (2014) during the 1950-1973 period have been excluded from this analysis. FAO data after 1974 are accepted as valid (Luckhurst et al., 2003) and show a 61 mt peak in 1976, with 27 mt annual catch during 1979-1990, and a lower average (15 mt) during 1991-1998 including a 3 mt minimum in 1991.

Fishery evolution and perspective

Low catches during the 1990s, followed a ban on trap fishing in 1990, to curb finfish overfishing. During this time, a lobster-specific trap was designed and tested, and a comprehensive management plan for the spiny lobster fishery was developed that included limits on input measures such as numbers of traps and fishing licenses (2).

Beginning in 1979, FAO reports started listing *P. guttatus*, locally known as "spotted spiny lobster" or "Guinea Chick lobster", as *Panulirus spp.*, but it had been harvested commercially since 1960 (Luckhurst et al., 2002), although from 1990-1998 it was not fished and only 3-4 fishers participated in the experimental fishery from 1998-2007. Variations in management and fishing effort affected *P. guttatus* landings in numbers, from 59% of total spiny lobster caught in 1975 (Luckhurst & Ward, 1996) to 27% in 1998 (3), and from 36% in weight (12 mt) in 1986, to 3% (1 mt) in 2006 (FAO, 2014), with annual landings relatively stable (about 3 mt) and equivalent to 10% of the spiny lobster fishery total weight, since 2008, when the current limited entry fishery began. From 1974-2015 Bermuda's spiny lobster fishery harvested a total of 1321 mt, including *P. argus* (1142 mt or 86%) and *P. guttatus* (179 mt or 14%), with most recent annual yields of 57 and 5 kg/km^2, respectively. During 1984-1997 a *P. argus* recreational fishery also existed, with catches in number of 2% (1984), 35% (1991) to 0% (1997) of total commercial catches, for an 11% period overall average (FAO, 2001). Although without catches data, 570 recreational permits were issued during the 2005/06 season, each recreational diver using a wire noose and allowed a daily bag of 2 spiny lobster from September 1st to March 31st (4).

Combining 1975-1991 catches in numbers and weight (FAO, 2001), it was calculated that *P. argus* individual weight ranged from 1.91 kg (1976) to 0.32 kg (1991 & 1996), averaging 0.84 kg, and that of *P. guttatus* ranged from 0.31 kg (1982) to 0.50 kg (1990) averaging 0.35 kg indicating that, individually, *P. argus* was 2.4 times heavier than *P. guttatus.*

Current fishery effort consists of 29 licensed fishers renting 348 government owned *P. argus* traps (12 per fisher); and 8 licensed fishers renting 144 *P. guttatus* traps (18 per fisher) harvesting 32 mt of *P. argus*, and 3 mt of *P. guttatus* (9% of total) in 2015 (5), with average daily outputs of 0.6 and 0.14 kg/trap, respectively, indicating that *P. argus* is roughly 4 times more abundant than *P. guttatus* in this very well managed fishery whose survival is most likely solely dependent on post-larvae from the Caribbean larval pool carried to Bermuda by the Gulf Stream.

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(2) Simons, D. 2014. Bermuda's fishery past, present and future. Master Maritime Std., Fish. Res. Magment., St. John's, Newfoundland, Major Report, 86*p.* [*Available from* http://research.library.mun.ca/6480/].

(3) "See footnote 2"

(4) "The Bermudian" (2014) (http://www.thebermudian.com/food-drink/1399-lobsters-in-bermuda1)

(5) "See footnote 1" (page 4)

Caribbean Basin spiny lobsters fisheries

General evolution

It is known that all Caribbean spiny lobster fisheries grew and peaked as early as 1957 (Turks & Caicos) or as late as 2011 (St. Kitts & Nevis, and Trinidad), to start declining thereafter with different rates and outcomes. Collectively *P. argus* fisheries grew 12 fold from 2957 tons in 1950 to a maximum of 36,696 tons in 1999, but experienced a 30% decline (to 25,557 tons) in 2013, for a 1.43 M mt total landed from 1950 to 2013 (FAO, 2014). These FAO figures are sub-estimates for all 3 spiny lobster species have been harvested since the early 1900s on every existing continental or insular shelf, but only 31 governments reported catches with different reliability levels during 16 to 80 consecutive year periods (Tables 1 to 4).

The fishing areas exploited by 5 Caribbean countries' fishers (Bahamas, Cuba, Florida, Honduras, and Nicaragua), account for 79% of the whole Caribbean's shelves area, and 86% of the total 1.43 M tons *P. argus* landed from 1950 to 2013 (FAO, 2014), are described separately, following.

The Bahamas fishery

Foreign fishing activities, and poaching

Bahamas is a widely disperse archipelago of several hundred islands, cays , and islets collectively designated as the "Lucayan Archipelago" with 26 larger populated islands in a discontinuous shelf with its largest portion called the "Great Bahamas Bank". FAO (2014) report that The Bahamas, harvesting only *P. argus* on a 116,500 km^2 shelf, landed from 1955-2013 a total of 263,320 tons, and in 2000 became the largest *P. argus* producer with 8457 annual harvests; both being sub-estimated because other countries harvested *P. argus* in Bahamian territorial waters during different periods. Florida east coast fishers harvested 22,700 mt during 1964-1979 (Labisky et al., 1980); some Cuban northeast coast fishers harvested 755 tons from 1969-1972, south of Andros Island (Millares and López, 1974). Both Floridian and Cuban fishers ceased their Bahamian activities before Bahamas commenced enforcing its territorial waters exclusive fishing rights in 1980, but fishers from the Dominican Republic pouched a total of 12,348 tons from Bahamas territorial waters from 2002-2014 (FAO, 2015) (6) (7).

Corrected landings and larval supply

Adding 35,803 tons harvested by other countries in Bahamian territorial waters brings the 1955-2013 total to 299,123 tons, representing a 2.6 tons/km^2 total yield for the period. Out of 30 sites within the Bahamas included in larval connectivity studies by Kough et al. (2013), 25 are theoretically considered as larvae importers, coinciding with a previous conclusion that less than 20% of Bahamas native larvae are retained (8), although more recently Callwood et al. (2016) concluded that "*Despite these international connections, the lobster populations within The*

(6) Smith, L. 2011. The theft of Bahamian fishery resources. 4 *p.* (Available from http://www.bahamapundit.com/2011/01/the-theft-of-our-fishery-resources.html)

(7) Sullivan-Scaley, K. M. 2011. Assessment of IUU (Illegal, Unreported and Unregulated) fishing in the Bahamas spiny lobster fishery- Univ. Miami, Final Report. 49*p.* [Available from http://www.academia.edu/10657835/FINALREPORTAssessment_of_IUUIllegalUnreportedand_Unregulated_Fishing_in_the_Bahamian_Spiny_Lobster_Fishery].

(8) Callwood, K.A. 2010. Use of larval connectivity mostly to determine settlement habitats of *Panulirus argus* in the Bahamas as a pre-cursor to marine protected area network planning. *Univ. Miami, Scholarly Repository- electronic Theses and Dissertations*. [Available from http://scholarlyrepository.miami.edu/oa_theses].

Bahamas have a high probability for self-recruitment, signifying domestic connectivity." again underscoring how difficult determining actual larvae supply origins to any given fishing area is.

Cuba fishery

Fishery characteristics

Cuba, the largest of the Greater Antilles, in reality is an archipelago with hundreds of islands, cays, and islets surrounding a main island with 4 shelves separated by depths in excess of 1000 meters, with the 2 southern shelves being relatively wide, and the 2 northern shelves being narrower. All 3 of the spiny lobster species have been reported but only *P. argus* is commercially harvested, with different yields per shelf suggesting the presence of individual populations. To investigate this assumption a total of 17,884 individuals were tagged in 1964 with 5203 being recaptured, always within the shelf they were tagged demonstrating that each shelf indeed sustained an isolated population (Buesa, 1965). Among all Caribbean *P. argus* producers, Total landings of 500,825 mt during 80 years (1935-2014) negatively affected the populations with varying degrees causing that, from being the largest producer since 1954, by 2000 was overtaken by The Bahamas in that position.

Cuba's four fishing areas evolution and present situation

Southeastern shelf landings peaked in 1983 (2976 mt) decreasing to 600 mt (80% reduction) in 2014, with fishing effort (annual fishing days) varying from 10,667 days in 1983, and 12,750 days in 1999, to about 3000 days in 2010 (Alzugaray and Puga, 2012). Southwestern shelf's landings decreased 65%, from 8523 mt in 1985, to 2961 mt in 2014, due to its spiny lobster population reduction from 122 million individuals, to 77 million during that period (Alzugaray et al., 2016); the effort also declined 36%, from 28,000 fishing days in 1985, to about 18,000 days in 2001 (FAO, 2003). The northwestern shelf is the smallest, and peaked sooner (1969) with 706 mt, declining 76% in 2015 when only 151 mt were reported, resulting from a halved population biomass (5500 mt to 2800 mt), and a 71% fishing effort reduction (2935 to 858 fishing days) both from 1955-2015 (Pintueles at al., 2016). Northeastern shelf's landings peaked in 1987 (2730 mt) decreasing 64% (to 973 mt) in 2014; the fishing effort peaked in 1999 (about 12,300 fishing days) decreasing 19% (to 10,000 fishing days) in 2001 (FAO, 2003). All those calculated biomasses and populations reductions are symptoms of a declining fishery in need of management intervention to assure its survival and potential recovery.

Present situation causes

Excessive fishing effort, driven by the international commercialization of spiny lobsters as a hard currency source for the Cuban government, is the fundamental cause for this fishery decline. Despite frequent official statements about a stringent implementation of all protective measures, the government approved closed seasons waivers during the so-called "Special Period in Peacetime" when economic assistance to Cuba stopped subsequent to the Soviet Union's collapse. This situation was described by Muñoz-Núñez [9] when she wrote:

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(9) Muñoz-Núñez, D. 2009. The Caribbean spiny lobster fishery in Cuba. An approach to sustainable fishery management. *MS of Environmental Management. Nicholas School of the Environment. Duke Univ., Final Report*, 97 p.

"*Because of the high market value of the lobster, an increase in its production (by fishing during the entire year) may have been considered as a quick solution to obtain revenues in order to support the ailing national economy"*. This decision resulted in 33 consecutive months (June 1998 to February 2001) of spiny lobster open season resulting from not implementing the 1999 and 2000 closed seasons determining fishing efforts historical maxima in 1999 for the southeastern (Alzugaray and Puga, 2012), and northeastern (FAO, 2003) shelves. Waiving 2 consecutive closed seasons was a very unwise decision and seems to have been the fishery's "*coup de grace*" for 1991-1998 annual landings averaged 9326 mt, while the annual average for 2001-2014, after the 1999-2000 "special period", have been 5416 mt only, representing a 42% reduction. Spiny lobster abundance decline following those 33 months of continuous harvest, along with fishing effort decreases enacted as part of some sort of "population recuperation" plan, resulted in even further landings reductions. Expected operation costs increases due to declining populations surely did not play a role in effort reductions, for since 1960 the Cuban spiny lobster fishery has been considered as a hard currency source and, as such, sales in the international market of the whole harvest to exchange an internationally worthless Cuban currency for US dollars or Euros has been a long held government policy.

Additionally to population decline driven causes, recently Cobas et al. (2012) calculated that 3254 mt of the total 23,242 mt harvested during the 1996-2014 in the northeastern shelf (14% of that total) were illegally harvested, and that this total was equivalent to 35% of all illegal catches nationwide. Under such assumptions, national illegal catch was 9297 mt, or almost 8% of the 121,000 mt landed during 1996-2014. Unquantified amounts of illegal harvests had been reported earlier by Muñoz-Núñez (*"While in my research I did not find any records of the number of illegally harvested lobsters, illegal fishing is a common activity in Cuba"*) (10), these facts being proof of regulations enforcement not as stringent as reported to be.

Total yields (t/km^2) for the 1957-2014 period varied by shelf, from about 17 t/km^2 in the 2 southern shelves, to 4 t/km^2 in the northwestern, and 14 t/km^2 in the northeastern shelves. For the 80 years 1935-2014 period the overall total yield was 15 t/km^2, the largest for the whole Caribbean basin.

One connectivity study (Kough et al., 2013) theorizes that Cuba's 2 southern shelves are larvae exporters, while the whole northwestern shelf imports larvae, and that 10 out of 14 sites in the northeastern shelf, import larvae, whereas 4 sites exported larvae. Other connectivity studies estimate that the southwestern shelf is self-sustained (Cruz et al., 2015); while other (11) calculated it imports 25% of post-larvae, and in Gutiérrez et al., (2012) that number is elevated to 43%, adding that larvae from the southwestern shelf reach other areas at different rates: Gulf of Mexico (62%); northeastern Florida (33%); northwest Cuba (3%), Yucatán (2%), and south Florida (0.2%).

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(10) "See footnote 9" (page 7)

(11) Gutiérrez, A. R. 2012. [Dispersion and chaoticity of passive particles in ocean waters off the western region of Cuba] - *Cuba, Oceanology Inst, & Meteorology Inst. (Ph.D.Theses) 212p.* [In Spanish]

Table 1 - Totals and annual averages during different periods (1936-2015)

Country or area (period) (spiny lobster species)	Shelf surface (km²)	Period data		Annual averages		e or i	u; d; m
		total landings (mt)	total years	t/year	kg/km^2		
Anguilla (1974-2013) (*P.a.* + *P.g.*)	3392	4856	40	121	36	e	u
Antigua & Barbuda (1961-2014)	3568	8170	54	151	42	e	u
Bahamas (1955-2013)	116,550	299,123	59	5070	43	i	d
Belize (1946-2014)	6940	39,520	69	573	83	e	d
Bermuda (1974-2015) (*P.a.* + *P.g.*)	615	1321	42	31	51	i	d
British Virgin Islands (1961-2013)	755	2656	53	50	66	i	u
Colombia (La Guajira) (1950-2013)	1255	7564	64	118	94	i	m
Costa Rica (1957-2013)	2400	9423	57	165	69	i	m
Cuba - SE shelf (1957-2014)	5702	95,058	58	1638	287	e	d
Cuba - SW shelf (1957-2014)	17,490	297,084	58	5122	293	e	d
Cuba - NW shelf (1955-2015)	3629	15,930	61	268	72	i	d
Cuba - NE shelf (1957-2014)	5624	78,146	58	1347	240	i	d
Dominican Republic (1968-1996)	8000	21,987	29	758	95	e	m
Florida (1936-2015)	13,000	154,317	80	1930	148	i	d
Grenada (1978-2013)	1595	649	36	18	11	e	u
Guadeloupe (2010-2012)	2110	278	3	93	44	e	u
Haiti (1954-2014)	5857	25,260	61	414	71	i	m
Honduras (1954-2013) (*P.a.* + *P.g.*)	69,370	154,655	60	2578	37	i	m
Jamaica (1968-2013)	3420	1596	46	35	10	i	m
Pedro Bank (1968-2013)	4170	6382	46	139	33	i	m
Martinique (1950-2014) (*P.a.* + *P.g.*)	1270	7835	65	121	95	e	u
México (Caribbean) (1964-2014)	12,000	36,976	51	725	60	i	d
Montserrat (1993-2010) (*P.a.* + *P.g.*)	157	3	18	0.2	1	e	u
Nicaragua (1963-2014) (*P.a.* + *P.g.*)	53,530	132,766	52	2553	48	e	m
Panamá (1950-2013)	12,690	13,480	64	211	17	i	m
Puerto Rico (1969-2013)	4073	6198	45	131	34	i	u
Saba Bank/St. Eustatius (1981-2014)	2200	514	34	15	7	i	u
St. Kitts & Nevis (1994-2013)	770	544	20	27	35	i	u
St Lucie (1990-2009)	522	352	20	18	34	e	u
St. Vincent / Grenadines (1994-2013)	1800	205	20	10	6	e	u
Trinidad (1998-2013) (*P.a.* + *P.g.*)	20,400	341	16	21	1	e	u
Turks & Caicos (1952-2013)	6500	16,802	62	271	42	i	d
US Virgin Islands (1974-2013)	1635	12,771	40	319	195	i	u
Venezuela (1950-2010)	1250	14,535	61	238	191	e	m
BRAZIL (1950-2013) (*P.a.* + *P.l.*)	75,000	394,001	64	6156	82	ss	ss

"e" = larvae exporter "i" = larvae importer "ss" = self-sustained

"d" = downstream area "u" = upstream area "m" = Caribbean middle areas

P.a. = *Panulirus argus* *P.g.* = *P. guttatus* *P. l.* = *P. laevicauda*

Gutiérrez et al. (2009) also calculated that larva from Cayman Islands will go to Jamaica (57%); Hispaniola (20%); northwest Cuba (10%); northeastern Florida (8%); southeastern Cuba (2%), and south Florida (1%) the only caveat being that, given Cayman Islands' spiny lobster population's small size, those post-larvae numbers most likely will be minuscule. Other theoretical dispersion models present Caribbean post-larval arriving Cuba from Honduras, and Belize (Cowen et al., 2002); and from Mexico, and Haiti (Chávez and Chávez - Hidalgo, 2012).

Colombia, Honduras, and Nicaragua fisheries

Corrected landings

FAO (2014) reports Honduras' spiny lobster landings (essentially *P. argus*) as 138,881 mt from 1954-2013, and Nicaragua's as 117,278 mt during 1963-2014 but both totals are underestimates. Along with 5124 mt from Colombia, all were harvested from the same mostly Nicaraguan continuous Caribbean continental shelf, which was the source of several fishing rights disputes between Nicaragua and its neighboring countries. This fishing area, with a surface of 122,900 km^2, also includes 6 oceanic banks, several close by isles and reefs, all exploited by these 3 countries during different periods causing additional fishing rights litigations.

Data from 1980-1977 corresponding to the collective exploitation of that "shared" shelf (FAO, 2001) permit adding 15,774 mt to Honduras' landings, for 154,655 mt total. Nicaragua's total was increased to 132,766 mt, when 15,488 mt, including 5124 mt harvested by Colombia in the disputed areas, were added. Colombia's total landings of 12,688 mt during 1950-2013 (FAO, 2014) include 7564 mt harvested in its territorial waters (in "La Guajira" area) (Prada et al., 2005), and the rest (5124 mt) were harvested in the disputed banks and isles near Nicaragua and were reassigned to the latter, reducing Colombia's total to the 7564 mt harvested in its territorial waters (Table 1).

Recent yields and juveniles illegal harvests

Exploiting common areas, Nicaragua's and Honduras' 2.2 and 2.4 mt/km^2 respective yields, include unknown quantities *P. guttatus* also harvested in these disputed areas by these 3 countries. FAO (2001) report that 24.29 M juvenile spiny lobsters, weighing 2643 mt and equivalent to 12% of the total 1996-2001 Honduras's harvest, were commercialized; also stating that up to 60% of landings in some areas, are juveniles. Nicaragua's situation is not better for around 30% of landings (18,720 mt) were juveniles (55.06 million) during 1990-2007 (FAO, 2007). Ehrhardt (2005) calculated that during the early 2000s, juveniles exceeded 40% of total annual landings (about 1700 mt), something FAO had previously reported (2001) with a calculated number of about 3.06 M juveniles caught annually.

Larval connectivity

Connectivity calculations (Kough et al., 2013) describe all areas close to Nicaragua and at the Nicaragua-Honduras frontier, as larval exporters, while the nearby oceanic banks and isles, are described as importers; the shelf close to the Honduras-Belize frontier is also considered as importer. Other theoretical larval connectivity study (Cowen et al., 2002) describe Honduras, and Belize exporting larvae to Cuba.

Table 2 - Total and annual average catches during the growing and after peaking periods (1935-2015). Unless specified, they are *P. argus* catches

Country or area (from first year with data)	Total catch (mt)	growing period			after peak to most recent		
		catches (mt)	years	average annual catch	catches (mt)	years	average annual catch
Anguilla (1974-2013)	4856	4203	35	120	653	5	131
Antigua & Barbuda (1961-2014)	8170	4508	38	119	3662	16	229
Bahamas (1955-2013)	299,123	194,451	49	3968	104,672	10	10,467
Belize (1946-2014)	39,520	15,649	36	435	23,871	33	723
Bermuda (1974-2015) *P. argus*	1142	144	3	48	998	39	26
Bermuda (1976-2015) *P. guttatus*	179	96	11	9	83	22	4
British Virgin Islands (1961-2013)	2656	2086	35	60	570	11	52
Colombia (La Guajira) (1950-2013)	7564	5760	40	144	1804	21	86
Costa Rica (1957-2013)	9423	2300	6	383	7123	48	148
Cuba (1935-2014)	500,825	277,604	51	5443	223,221	29	7697
Dominican Republic (1968-1996)	21,987	4965	18	276	17,022	26	655
Florida (1936-2015)	154,317	110,672	61	1814	43,645	19	2297
Grenada (1978-2013)	649	337	21	16	312	14	22
Guadeloupe (2010-2012)	278	198	2	99	80	1	80
Haiti (1954-2014)	25,260	13,010	43	303	12,250	18	681
Honduras (1954-2013) (*)	154,655	61,551	33	1865	93,104	27	3448
Jamaica (1968-2013)	7978	6380	32	199	1598	8	200
Martinique (1950-2014)	7835	2700	10	270	5135	49	105
México (Yucatán) (1955-2014)	36,976	28,918	48	602	8058	12	672
Montserrat (1993-2010)	3	2	13	0.15	1	5	0.20
Nicaragua (1963-2014) (*)	132,766	70,062	38	1918	54,780	15	3652
Panamá (1950-2013)	13,480	11,054	32	345	2426	8	303
Puerto Rico (1969-2013)	6198	1756	11	160	4442	34	139
Saba Bank/St. Eustatius (1981-2014)	514	241	6	40	273	5	55
St. Kitts & Nevis (1994-2013)	544	508	18	28	36	2	18
St. Lucie (1990-2009)	352	249	12	21	103	8	13
St. Vincent / Grenadines (1994-2013)	205	63	2	32	142	9	16
Trinidad & Tobago (1998-2013)	341	274	14	20	67	2	34
Turks & Caicos (1952-2013)	16,802	5100	22	232	11,702	40	293
US Virgin Islands (1974-2013)	12,771	5295	18	294	7476	22	340
Venezuela (1950-2010)	14,535	9015	40	225	5520	20	276
CARIBBEAN TOTAL (1950-2013)	1’439,553	1’036,359	50	20,727	403,194	14	28,800
BRAZIL (1950-2013) *P. argus*	352,561	191,325	42	4555	161,236	22	7329
BRAZIL (1970-2013) *P. laevicauda*	41,440	18,685	9	2076	22,714	35	649

(*) Harvested in the same continental shelf

Florida fishery

Evolution and corrected landings

Florida's shelf is included within the 3 *Panulirus* species' ranges (Holthuis, 1991) but only *P. argus* is reported by Nizinski (2003). Excluding 22,700 mt that, according with Labisky et al. (1980) were harvested in the Bahamas (1964-1979) and are reassigned to that country's total (Table 1), Florida's harvests from 1936-2015 total 154,317 metric tons. Landings reported from North and South Carolinas, Alabama, Louisiana, Mississippi, and Texas from 1950-2015 amount to 553.3 mt only, making it an almost exclusive Florida fishery, with 96% harvested in its southern fishing areas (SEDAR 8, 2010). During the 1895-2014 period landings increased 34 fold, and prices almost 3000 fold; and while traps numbers from 1927 to present increased by a factor of 900, the annual catch per trap decreased 24 fold. Commercial landings peaked in 1996 (3569 tons), with 2690 tons average harvests during the 19 years before 1996, which is 18% more than the 2297 tons average harvest during 19 years after the peak harvest (Table 2).

Fishery characteristics and yields

Total effort, measured as trap licenses numbers, was largest in 1991 with 0.94 million which coincided with the lowest gross CPUE (Catch Per Unit Effort) yet (1.6 kg/trap*year). This fact, along with operational problems caused by such very large number of traps used, determined the implementation of a Trap Reduction Program (TRP) in 1993, aimed at reducing licensed traps to a still not reached goal of only 0.4 million. After the TRP started CPUE increased 3.7 times until the 1999/00 fishing season when it decreased 19%, the same value as in 1994/95 when traps were 40% more numerous, indicating a population declining trend warranting further effort reductions to, at least, the 0.4 million traps initialed aimed by the TRP.

Florida *P. argus* fishers require a license and each trap is registered making it a "limited access" and well-regulated fishery but is negatively impacted by 2 fundamental problems: a "recreational" fishery, and using immature-sublegal size animals, called "shorts", as attractants in the trap fishery. The "recreational" fishery average 22% of the total harvest or about 700 mt, but that contribution is considered to be an underestimate for actual participant numbers and their catches are obtained through a voluntarily answered, unenforceable or independently corroborated survey (SEDAR 8, 2010) (Figure 1)

Juveniles use and larval supply

Using sub-legal individuals (pre-recruits) as attractants caused the death of 28.32 million juveniles with 9629 mt total weight from 1978/79 to 2008/09, equivalent to 16% the total harvest during that period (SEDAR 8, 2010). Besides this juvenile authorized use, illegal harvests and undersize lobsters sales was recognized as a continuous problem in the early 2000s (GMFMC, 2002) both practices reducing potential recruiters numbers to the fishery, affecting also the population reproductive ability even if locally produced larvae are essentially "*expatriated out*" (Yeung and Lee, 2002). Only 1 of 8 Florida sites are considered as larvae exporters, while the rest are larvae importers (Kough et al., 2013); other theoretical studies calculate that 23% of larvae produced in Cuba are recruited to Florida (Gutiérrez et al., 2012), along with larvae from the Dominican Republic, and Mexico (Chávez and Chávez - Hidalgo, 2012); Honduras, and Belize (Cowen et al., 2002). Due to it "downstream" geographical position, during the 80 years 1936-2015 period, this fishery shows an 11.9 t/km^2 yield, close to Cuba's, and almost 5 times larger than Bahamas, Nicaragua, and Honduras yield averages.

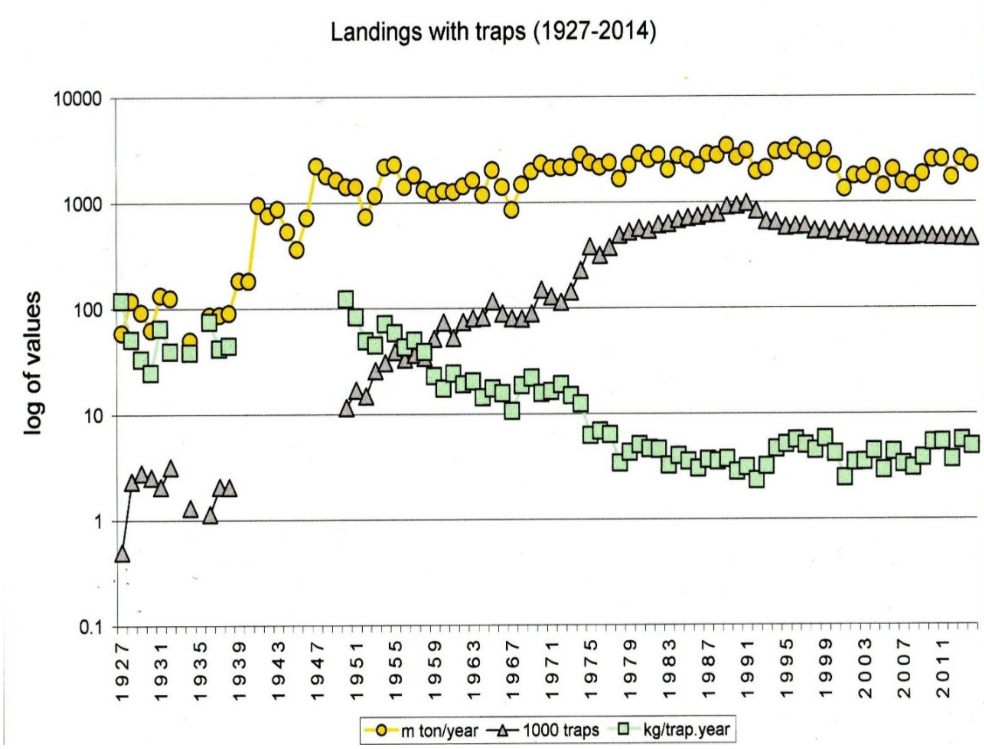

Figure 1 - Florida trap fishery indexes variations: Catch (59 to 2231 mt); effort (499 to 455,212 traps); CPUE (118 to 5 kg/trap*year)

Caribbean remaining spiny lobster fishing areas

The remaining fishing areas represent 21% of the combined Caribbean shelves surfaces and 16% of regional landings, with Belize and Yucatán peninsula (México) being more relevant. Belize harvested 39,520 mt during 1946-2014 (17% of the 16% collective total harvest) in a 6940 km^2 shelf, for a 5.7 t/km^2 yield during the period, with 3 out of 10 sites theoretically importing and the rest exporting larvae (Kough et al., 2013). Although actual numbers are unknown, it is suspected that large quantities of juveniles are commercialized (FAO, 2001). Yucatán fishers, mostly based in Yucatán, and Quintana Roo states (México) harvested 36,976 mt on a 12,000 km^2 shelf, for a 3.1 t/km^2 yield (1955-2014); with 5 out of 10 sites in the regional connectivity study theoretically importing post-larvae, and the other 5 exporting them (Kough et al., 2013).

Juvenile commercialization throughout the Caribbean

Special fishery regulations have always existed, be it closed seasons, size limits, protected areas, forbidden gears, and most commonly to all areas, berried females catch and commercialization prohibition (McManus and Lacambra, 2012) but also since their inception, the most damaging regulations violations have been known to be juveniles harvest and commercialization. Fisheries violations reports exist for seven of 23 producers, which does not mean they are the only subjecting their spiny lobsters populations to unsound practices. From 1992-2004 a total of 94 fishery violations, including undersized and lobsters possession, and berried females were recorded in Antigua & Barbuda, with only 2% of all operators fined in 1992/94, and 3% in 2000/01 (FAO, 2007).

Juveniles' commercialization is perhaps the most common violation and in Costa Rica has been reported to be about 30 mt or 44% of the total 2004 catch (Wehrtmann, 2004). In the Dominican Republic around 70% of lobsters caught in "Parque Jaragua" in 1997 were below legal size (FAO, 2001). In Jamaica 30% of 221 mt landed in 2005, equivalent to 0.2 million juveniles, were commercialized, and in 2007, 76% of 228 mt (0.67 million) lobsters were immature females, with an overall estimate of 0.87 million juveniles caught annually (CARICOM, 2007). In Puerto Rico juveniles were 40% of 174 mt (0.2 million) landed during 1985-1989, which increased to 59% of 107 mt (0.19 million) during 1989-1991, followed by a reduction to 24% of 48 mt (0.03 million) in 1998, for a calculated 0.14 million juveniles harvested annually (SEDAR 8, 2005). In the US Virgin Islands, probably because it is a smaller and easier to control fishery, only 1.3% of St. Croix's landings in 1987-1989, and 2.9% of St. Thomas' and St. John's landings in 1985-1989, were juveniles (SEDAR 8, 2005a.). Juvenile spiny lobsters deaths in St. Lucia result from being used as attractants in traps, and from being caught towards the closed season's end and "stockpiled" for many days in large shallow water enclosures waiting to be legally commercialized (FAO, 2003). Juveniles preference by restaurants' owners add "market pressure" to their harvest (FAO, 2001), determining that about 20-62% of the 2 to 12 mt St. Lucia's annual harvests are juvenile lobsters.

All regional mentioned violations are fully known in each corresponding country, their persistence being sign of an official unwillingness or inability to eradicate these problems. Along with the hundreds of thousands juveniles Florida fishers "legally" use as attractants in their traps, about 25% of all Caribbean harvested *P. argus* are pre-recruits, unable to reproduce and contribute to the collective larval pool.

Brazilian spiny lobsters fisheries

Brazilian *P. argus* subspecies

Based on mitochondrial DNA (mtDNA) differences, Sarver et al. (1998) proposed the existence of two *P. argus* subspecies, the "*argus argus*" inhabiting the Caribbean Basin at large, and the "*argus westonii*" in Brazilian fishing areas, some of which have been found in Venezuela, and Florida (Sarver et al., 2000) probably consequence of a larval "spill-over" from Brazilian spiny lobster populations into the Caribbean, such larval transference also explaining the *P. laevicauda* declining abundance from east to west along the Caribbean, described earlier. To date there are no anatomic or taxonomic descriptions for these two proposed subspecies, nor will it probably ever be for mtDNA differences, most likely caused by *P. argus westonii* genetic isolation, not necessarily produce differentiating phenotypic traits. Sarver et al. (1998) described "*characteristic color patterns*" distinguishing both subspecies but color differences in *P. argus* hardly can be of diagnostic taxonomic value for this species is known to mimic bottom color hues from the area last molted or inhabited in, which is considered as a defense mechanism (Smith, 1948). Genetic differences between Florida and Brazil *P. argus* have been also reported by Diniz et al., (2005a; 2005b) reaffirming genetic differences between a Brazilian *P. argus* autochthonous and isolated population, and a Caribbean meta-population, but Nizinski (2003), referring to *P. argus argus* and *P. argus westoni* subspecies, wrote: "*until a formal revision is conducted, the taxonomic status of these subspecies remains uncertain"*.

Brazilian spiny lobsters population's origin

Brazilian spiny lobsters populations have been described as self-sustained, their larvae drifting in wide-circular gyres near the coast within the Equatorial and North Brazilian currents and counter currents system including several retro-flections and eddies (Cruz et al., 2015) that could account for the presence of a *P. argus* permanent population commercially exploited during the last 15 to 20 years in the Central-Eastern Atlantic Cabo Verde Archipelago (Freitas and Castro, 2005). Later, and contrary to this scenario, Góes et al. (2007) described a westward larval flow for this Central-Eastern Atlantic area instead, rejecting any eastward larval drifting pattern originated in Brazil.

Some argue that *P. argus* documented presence in two different West Africa Ivory Coast locations (Holthuis, 1991) could have originated in the Caribbean without a clear explanation of how such extremely lengthy, and seemingly quite improbable migratory path took place. Góes et al. (2007), while considering a transatlantic eastward migration originated in Brazil as impossible, accepts a westward spiny lobster larval connectivity from Ascension Island, S. Pedro & S. Paulo Archipelago, and from Atoll das Rocas & Fernando Noronha that could explain the presence of *P. echinatus* in Brazilian fishing areas. These conflicting conclusions leave the origin of *P. argus* populations in Cabo Verde archipelago unexplained, underscoring the inherent difficulty of calculating with probabilistic certitude any given planktonic theoretical larva connectivity pattern. The fact remains that, given the theoretical behavior analysis of the existing currents around Brazil (Góes et al., 2007; Cruz et al., 2015), an eastwardly counter current carrying Caribbean born larvae into Brazil is highly improbable which, added to existing genetic results defining a locally isolated subspecies, allows concluding that *P. argus* and *P. laevicauda* Brazilian populations are self-sustained.

Brazilian fishery evolution and present stock situation

Brazilian spiny lobster populations inhabit a 75,000 km^2 shelf where, during the 1950-2013 period, a total of 0.35 M mt or 23% of all *P. argus'* landings, and the totality of all *P. laevicauda* landings took place (FAO, 2014). Fishers use baited traps, bottom gillnets, diving with "hooka" or SCUBA on artificial shelters and "shades" to attract lobsters, this self-sustained fishery harvests *P. argus, P. laevicauda*, and *P. echinatus*, the latter in minuscule and mostly unreported amounts.

Historically, *P. argus* and *P. laevicauda* combined harvests peaked in 1991 with a 11,059 mt catch (FAO, 2014), and 23 years latter is still annually producing more than 6500 mt, with the Northern fishing area considered as fully exploited, and those in the Northeast as overexploited and facing a problematic long-term survival (Negreiros-Aragão, 2015). This situation is presently aggravated because of fishing conditions described as equivalent to an "open access" fishery (Santana and Cruz, 2016) the fishers generally disregarding existing biological, fishing gear, and access regulations, and only the closed season being enforced.

Recently the stock has been described as heavily exploited since the 1980s painting a pessimistic present day scenario maybe close to collapse, with a maximum sustainable yield (MSY) of only about 5000 mt, consistently surpassed annually during the last 15 years (Andrade, 2015). This MSY was calculated applying a biomass dynamic model based on a Baysian approach without assuming equilibrium, and using "tons/million pots x days" as CPUE, all of which is acceptable given the fact that the stock survival solely depends on its parental population's reproductive potential without foreign post-larval contributions (Cruz et al., 2015). The *P. argus* fishery most recent annual yield was 79 kg/km^2 (Table 4).

Situation of the *P. laevicauda* fishery

From 1970-2013 *P. laevicauda* harvests amounted to about 0.04 M mt or 10% of the combined almost 0.40 M mt total for both spiny lobsters fisheries (FAO, 2014) but presently it is described in a very precarious situation. Historically *P. laevicauda harvests* have represented varying proportions of total landings, from 40% in 1961-1969 (Buesa and Paiva, 1969) or at least about 13,000 additional tons during those years, they were a 36% in 1978, decreasing thereafter to only 10% in 1986 (a 75% decline in only 17 years), to no longer being even reported by FAO from 1986-2013 (FAO, 2014). Negreiros-Aragão (2015) calculated its catches are now only 5% of landings, and were not even mentioned in most recent harvests data off the State of Ceará (Santana and Cruz, 2016). Its declining harvests from 1961 onwards are likely indicators of a constant population biomass reduction probably because, being harvested in shallower waters (Negreiros-Aragão, 2015), has been more affected by an ever increasing and unregulated fishing effort pointing to a bleak panorama for this species. The *P. laevicauda* fishery most recent annual yield was 2 kg/km^2 (Table 4).

Brazilian spiny lobsters fishery's isolation and self-sustained condition requires strict enforcement of all its regulations, and developing a robust management program that should include fishing effort reduction, and determining annual recruit population sizes to prevent its collapse.

Discussion

Larval connectivity vs. fishery's geographical position

Within the Caribbean Basin larvae arrival to each fishing area is accepted to be the connectivity ultimately consequence, as summarized in Kough et al. "Figure 5" (2013), where they classify all Caribbean areas into "*larval exports and imports at a site (n = 261), after removing self-recruitment."*(Sic.). That classification permitted to identify the 34 Caribbean fishing areas into 15 "exporters", and 19 "importers" (Table 1) the assumption being that "exporters" produce excess larvae or their larvae are not retained, while importers receive additional larvae from other areas which should result in larger populations and greater yields, but when the "exporting" larvae areas' average yield (81 kg/km^2) is compared with the "importing" areas' average yield (72 kg/km^2), the difference is not significant ($P > 0.31$).

Table 3 - Average annual yields during different periods (1936-2015). Unless specified, these are *P. argus* catches yields.

Country or area	Shelf surface (km²)	average annual yield (kg/km^2)			
		before peak	at peak	after peak	most recent
Anguilla (1974-2013)	3392	35	68	39	40
Antigua & Barbuda (1961-2014)	156	35	112	73	49
Bahamas (1955-2013)	116,550	34	89	83	65
Belize (1946-2014)	6940	59	145	105	85
Bermuda (1974-2015) *P. argus*	615	67	99	40	57
Bermuda (1976-2015) *P. guttatus*		14	20	6	5
British Virgin Islands (1961-2013)	755	75	222	70	53
Colombia (La Guajira) (1950-2013)	1255	107	418	68	77
Costa Rica (1957-2013)	2400	133	292	63	5
Cuba - SE shelf (1957-2014)	5702	282	522	287	105
Cuba - SW shelf (1957-2014)	17,490	321	464	259	169
Cuba - NW shelf (1955-2015)	3629	81	195	67	42
Cuba - NE shelf (1957-2014)	5624	233	485	238	173
Dominica (2007-2010)	751	0.27	0.56	0.1	0.1
Dominican Republic (1968-1996)	8000	30	103	83	53
Florida (1936-2015)	13,000	137	275	176	207
Grenada (1978-2013)	1595	8	14	14	14
Guadeloupe (2010-2012)	2110	37	57	38	38
Haiti (1954-2014)	5857	49	162	119	77
Honduras (1954-2013)	69,370	24	104	48	24
Jamaica (1968-2013)	3420	16	43	11	18
Pedro Bank (1968-2013)	4170	51	141	46	57
Martinique (1950-2014)	1270	210	236	84	31
México (Caribbean) (1964-2014)	12,000	49	89	56	83
Montserrat (1993-2010)	157	0.9	1.9	1.3	1.3
Nicaragua (1963-2014)	53,530	34	68	68	68
Panamá (1950-2013)	12,690	25	83	19	57
Puerto Rico (1969-2013)	4073	37	57	32	24
Saba Bank/St. Eustatius (1981-2014)	2200	13	45	26	20
St. Kitts & Nevis (1994-2013)	770	35	62	27	19
St Lucie (1990-2009)	522	37	69	25	19
St. Vincent / Grenadines (1994-2013)	1800	14	21	8	16
Trinidad & Tobago (1998-2013)	20,400	0.6	6	2	1
Turks & Caicos (1952-2013)	6500	30	92	45	32
US Virgin Islands (1974-2013)	1635	140	85	208	44
Venezuela (1950-2010)	1250	162	901	229	72
BRAZIL (1950-2013) *P. argus*	75,000	114	147	98	79
BRAZIL (1970-2013) *P. laevicauda*		25	49	9	2
Correlation (*P*) between "yield" and shelf surface		< 0.03	< 0.005	< 0.005	< 0.004

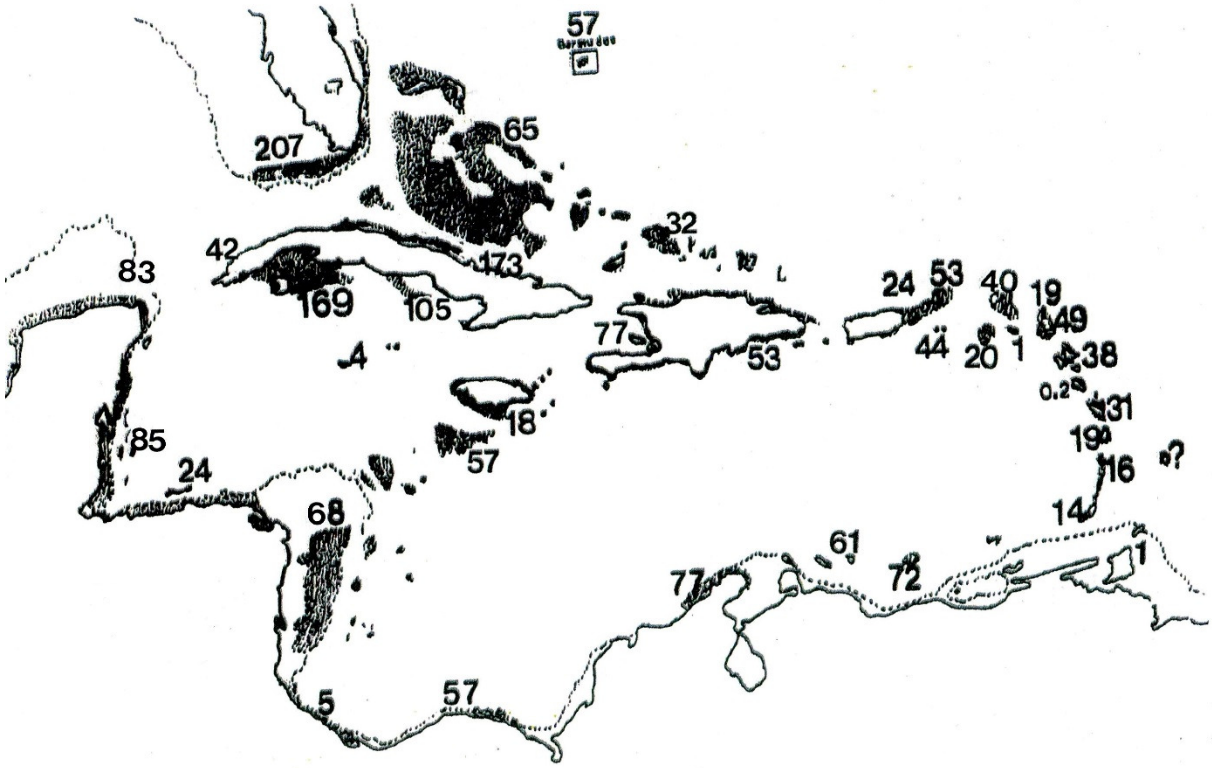

Figure 2 - Most recent yield values (kg/km^2) for Caribbean fishing areas (Table 3)

Considering only the overall Caribbean westward water circulation, those same 34 fishing areas (Table 1) were classified into 14 situated "upstream" (at the Caribbean eastern water "entrance"); 10 "downstream", at the "receiving end" of the general water flow, and another 10 in between ("middle" areas). Their respective average yields (43 kg/km^2 "upstream"; 130 kg/km^2 "downstream"; and 67 kg/km^2 for "middle" areas) were compared resulting that "downstream" yields were larger ($P < 0.015$), while the differences between "upstream" and "middle" yields were not, indicating that, from the practical (fishery) point of view, where the fishery is located within the Caribbean geography with relation to the overall westward regional water flow is key for the fishery's success. Also an "upstream" productivity average (78 t/km^2*year) 22 times smaller than the "downstream" fishing areas' productivity (1698 t/km^2*year) was statistically different ($P < 0.0002$). Given the positive significant correlation between shelf surface and harvest size with adequate effort and abundance, this was an expected result for the 10 "downstream" shelves' average surface areas (19,901 km^2) is 6.3 times larger than in the 14 "upstream" areas (3161 km^2). Larvae overall general westward flow is reflected in the sites' "connective" classification (Kough et al., 2013), for 70% of all "downstream" and "middle" fishing areas (14 out of 20) are "importers", while 64% (9 out of 14) of all "upstream" sites are "exporters". Consequently, and despite they may be considered as a simplistic approach, spiny lobster regional fisheries' commercial yields are better indicators of their present condition and fishing potential, than the most probable theoretical interaction between regional sea currents and imaginary particles, even in the most sophisticated connectivity models (Figure 2).

Caribbean connectivity of *P. argus* is not a unique phenomenon for Iacchei and Toonen (2013), using mitochondrial DNA (mtDNA), found connectivity for the Hawaiian spiny lobsters *P. marginatus* and *P. penicillatus* populations in 15 islands, atolls, reefs and banks along the Hawaiian archipelago's 2500 km extension. What is unique about *P. argus* Caribbean connectivity is its scale, determined by the geography and oceanography of this very large body of water definable as a "land surrounded section of the Western Central Atlantic Ocean" which, along with the Gulf of Mexico, has often been called the "American Mediterranean Sea" (Sverdrup et al., 1942). Its net westward water flow from the Lesser Antilles towards the Gulf of Mexico, the Florida Strait, and beyond, and a large cyclonic gyre comprising the Darién and Mosquitos Gulfs covering the Nicaraguan, Panamanian, and Colombian continental shelves, has resulted in *P. argus's* world record breaking harvests. It is also necessary noting that Caribbean larvae connectivity has to include *P. guttatus*, and *P. laevicauda* larvae as well, for such a huge water movement cannot influence *P. argus* larvae alone.

Present harvests commonalities

After an average of almost 50 years of continuous exploitation, there are many commonalities amongst all spiny lobster fisheries including all fishing areas' comprehensive knowledge, the absence of "professional secrets" about where and how to harvest, and the optimization of all harvesting gears and methods to each country's fishing areas characteristics despite of which, fisheries have declined at different rates. These known facts allow concluding that most recent harvests are solely dependent on each fishery's annual recruits abundance for all reserves have been substantially reduced, and reported harvests most likely are above "optimal" extraction levels in most of areas, especially where effort is not adequately. These exploitation levels were recognized by FAO (http://www.fao.org/newsroom/common/ecg/1000505/en /stocks.pdf) for all Western Central Atlantic (FAO area 31) *P. argus* populations, when they were considered either "fully exploited" or "overexploited", this perhaps being these fisheries' most important commonality.

Fisheries' evolution, productions, and yields levels

During the 1935-2015 period *P. argus*, and some *P. argus* with *P. guttatus* combined catches from 34 fishing areas show considerable differences, their totals and average annual yields both positively correlated ($P < 0.0001$) with their corresponding shelves' surfaces (Table 1) which is a logical expectation namely, that large shelves with adequate ecological conditions and enough post larvae supply should be able to sustain large populations rendering proportionally large harvests, if adequate effort is applied. Yields (kg/km^2), essentially dependent on population's size and fishing effort applied, are also positively correlated with total surfaces, but at different levels depending on each fishery's phase resulting in different harvest levels (t/km^2*year), and yields (kg/km^2) during each fishery's evolution phase (Table 3).

All fisheries show "growing" and "after peak to most recent" periods each lasting different years lapses essentially consequence of available information, with usually average annual catches larger in the latter period after effort already peaked. Table 2 summarizes these 2 periods for each of the 34 areas and fisheries with 26 years average for growing periods, and 19 years average during after peaking periods. The overall annual catches' average while growing (775 t/year) and after peaking (1296 t/year), although apparently different, are not

Table 4 - Spiny lobster catches from the peak year to most recent (1957-2015). Unless specified, data refer to *P. argus* catches

Country or area (peak year to most recent)	catch in metric tons		Years in the period	% catch decrease		most recent annual yield (kg/km^2)
	peak catch	most recent		in the period	annual	
Anguilla (2008-2013)	232	135	5	-42	-8	40
Antigua & Barbuda (1998-2014)	357	156	15	-58	-4	49
Bahamas (2003-2013)	10,378	7588	10	-27	-3	65
Belize (1981-2014)	1010	597	34	-41	-1	85
Bermuda (1976-2015) *P. argus*	61	35	40	-43	-1	57
Bermuda (1986-2015) *P. guttatus*	12	3	21	-75	-4	5
British Virgin Islands (2002-2013)	168	40	11	-76	-7	53
Colombia (La Guajira) (2002-2013)	524	97	21	-81	-4	77
Costa Rica (1962-2013)	700	13	52	-98	-2	5
Cuba - SE shelf (1983-2014)	2976	600	32	-80	-3	105
Cuba - SW shelf (1985-2014)	8123	2961	30	-64	-2	169
Cuba - NW shelf (1969-2015)	706	151	47	-76	-2	42
Cuba - NE shelf (1987-2014)	2730	973	28	-64	-2	173
Dominica (2009-2010)	0.42	0.07	1	-83	-83	0.1
Dominican Republic (1987-1996)	823	420	9	-49	-5	53
Florida (1996-2015)	3569	2690	20	-25	-1	207
Grenada (1999-2013)	73	23	14	-68	-5	14
Guadeloupe (2010-2012)	120	80	2	-33	-17	38
Haiti (1996-2014)	950	450	17	-53	-3	77
Honduras (1986-2013)	7180	1657	27	-77	-3	24
Jamaica (2005-2013)	148	60	8	-59	-7	18
Pedro Bank (2005-2013)	590	240	8	-59	-7	57
Martinique (1965-2014)	300	40	49	-87	-2	31
México (Yucatán) (2002-2014)	1070	1000	12	-7	-1	83
Montserrat (2005-2010)	0.26	0.21	6	-19	-3	1
Nicaragua (2000-2014)	6180	3629	14	-41	-3	68
Panamá (2005-2013)	1053	723	8	-31	-4	57
Puerto Rico (1979-2013)	232	98	34	-58	-2	24
Saba Bank/St. Eustatius (1999-2014)	100	43	14	-57	-4	20
St. Kitts & Nevis (2011-2013)	48	15	2	-69	-35	19
St Lucie (2001-2009)	36	10	9	-72	-8	19
St. Vincent / Grenadines (2004-2013)	38	29	9	-24	-3	16
Trinidad & Tobago (2011-2013)	126	21	2	-83	-42	1
Turks & Caicos (1957-2013)	5392	211	42	-65	-2	32
US Virgin Islands (2004-2013)	1405	72	23	-95	-4	44
Venezuela (1990-2010)	1126	90	20	-92	-5	72
CARIBBEAN TOTAL (1999-2013)	36,694	25,557	14	-30	-2	71
BRAZIL (1991-2013) *P. argus*	11,059	6198	23	-44	-2	79
BRAZIL (1978-2013) *P. laevicauda*	3639	124	36	-97	-3	2

($P > 0.88$) due to their large intrinsic variability, which does not invalidate the fact that, except for smaller mostly Lesser Antilles producers, Bermuda's *P. guttatus*, and Brazil's *P. laevicauda* fisheries after peaking annual catches are larger than during the growing period usually consequence of maximum fishing efforts near peak landings, and shorter time spans.

Average annual yields before (72 kg/km^2), after (76 kg/km^2), at peak (159 kg/km^2), and most recent (52 kg/km^2) for the 38 areas and fisheries (Table 3) are not statistically different [$F (3; 140) = 2.47$; $P > 0.06$] despite "peak" yields being numerically more than twice larger than during the growing period, and after peaking periods, all consequence of data intrinsic variability between areas. Caribbean annual yields average 31 kg/km^2 for Puerto Rico and Lesser Antilles; the Greater Antilles, Bahamas, and other nearby banks average 81 kg/km^2; Florida and continental countries average 80 kg/km^2; Bermuda's is 57 kg/km^2, and the entire Caribbean Basin's average annual yield is 62 kg/km^2 (Table 1; Figure 2).

When most recent landings, expressed as peak landings' percentage, are related with years from peak landings (Table 4; Figure 3) they show a significant negative correlation ($P < 0.006$), meaning that the more years elapsed after peaking, the smaller most recent landings are,

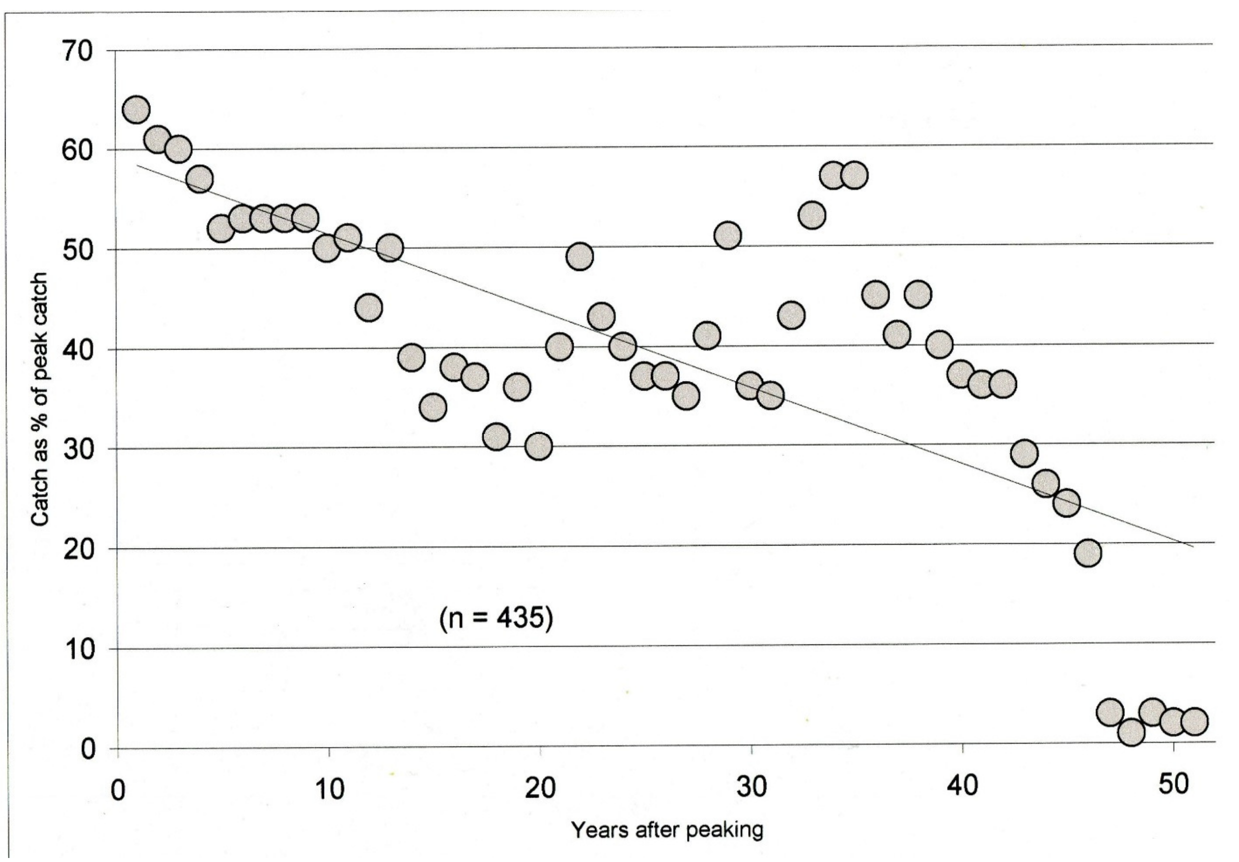

Figure 3 - Relation (P < 0.006) between catches as peak year percentages and years after peak catch

sign that either the populations, the fishing effort used, or both have declined. Within this general situation, post-peaking periods annual average percent decline in larvae importing areas (-4.9%) is not statistically different ($P > 0.37$) to that in larvae exporting areas (-6.7%), but annual average percent reduction in downstream fishing areas (-2.0%) is significantly smaller ($P < 0.02$) than for upstream areas (-10.3%), and when those in "middle" areas are included (-4.3%), the differences are statistically significant [$F(2;32) = 9.65$; $P < 0.0005$] because downstream areas annual reductions are actually smaller to those upstream ($P < 0.01$), and in middle areas ($P < 0.05$), again stressing the preeminent importance geographic position within the Caribbean basin has in the evolution of each of the considered fisheries, rather than the most probable behavior of imaginary particles under prevalent currents influences.

Fishing gears, effort, and CPUE

Diving (breath-holding snorkeling, with SCUBA, or using a compressor on board pumping air through a hose to a regulator in the diver's mouth, called "hooka") using harpoon, spear gun, snare, noose, or hook sticks are practiced in 41% of countries; pots or traps (square; "S" or "Z" shaped, with one or two entrances, made from different materials, baited with by-catch fish or salted cowhide) are used in 31%; nets (bully, gill, bottom, set, or trammel) in 20%; and artificial refuges or shades ("casitas" or "condos", and discarded vehicles' tires) to attract lobsters are used in the remaining 8% of the countries, while are forbidden in others (FAO, 2003; 2011; 2015).

Effort and CPUE information is disperse, non-reported for every fishing area and, most important, difficult to compare given the diversity of units used, many of which have little or no meaningful value to evaluate population abundances which is its essential objective. Although Harley et al. (2001) rejected such use to characterize cod and similar mixed demersal fisheries, and Maunder et al. (2006) arrived to a similar conclusion referring to mixed tunas pelagic fisheries, Saul and Die (2007), on the other hand, concluded that a "*standardized CPUE may be practical and lead to unbiased estimator of abundance for large-scale fisheries targeting a single species*" (Sic.) which is a fairly accurate description of any spiny lobster fishery, the remaining issue is the selection of an adequate CPUE that has to include a time component.

Different countries use different units (Table 5) some with questionable intrinsic value and only useful if applied consistently for the same fishing gear during many years. Some, such as "kg/trip" or "ton/ship" without defining gears used, their number, or trip duration, are of little value. Similarly, "kg/day", "kg/boat-day", "ton/boat-day", or "ton/boat-year" without data about the type and number of fishing gears, are also of limited value, especially if gears are "active" in nature. Belize's divers' activities, measured in "kg/fishers-day", would require data on the number of hours diving to obtain a better productivity value.

Passive fishing gears, such as traps, where catches depend on lobsters and traps combined "probability of encounter" (interaction), added to trap retention probability (percent retention rate), are the best to reflect lobsters abundances but even so, their value as abundance indicator depends on how CPUE is expressed. For instance "kg/trap-haul" (Antigua and Barbuda), or "lobsters/haul" (Bermuda) without knowing the number of days between hauls are not informative enough; and "kg/trap-year" (Florida, and Venezuela) or "lobsters/trap" (Bermuda) without the actual number of days used, are also deficient. The best CPUE found is "kg/trap-day" (or any other mass unit for that matter, or even number of lobsters of known average weight) used in Belize, Florida, and Brazil (Table 5) because it represents the weight of lobsters caught in each trap per day deployed, which intrinsically includes the trap retention rate

Table 5 - Catch Per Unit Effort (CPUE) and Effort evolution in some *P. argus* fisheries

Country or area		CPUE defined as	period with effort data	CPUE values during each period: peak value	end of period value	% less	with lowest effort	with peak effort	last to peak effort (%)	Ref.
Antigua & Barbuda		kg/trap-haul	2001-2011	0.95	0.46	-52	---	---	---	(1)
Bahamas (Andros)		kg/day	1969-1972	120	105	-12	120	105	same	(2)
Belize		kg/trap-day	1965-1997	1.5	0.6	-60	1.5	0.6	-10	(3)
		kg/fisher-day	1999-2009	3.1	1.9	-39	2.4	2.0	+6	(12)
Bermuda		lobsters/trap	1996-2001	7.6	4.3	-43	4.3	7.3	-28	(3)
		lobsters/haul		7.6	1.9	-75	2.0	7.3	-28	
Colombia (Luna Verde)		kg/boat-day	1993-1997	44	28	-36	32	30	+23	(4)
			1993-2001	204	79	-61	86	79	same	(3)
			1990-2002	500	315	-37	500	90	-32	(5)
Cuba - SE shelf		kg/day	1983-2010	345	190	-45	190	156	-76	(6)
SW shelf			1983-2001	316	234	-22	349	301	-31	(3)
NW shelf			1983-2015	154	68	-56	68	154	-71	(7)
NE shelf			1983-2001	228	108	-53	101	158	-18	(3)
Florida		kg/trap-year	1927-2015	6.1	5.7	-7	117.8	3.1	-52	(13)
		kg/trap-day	1985-2015	0.052	0.045	-13	0.038	0.028	-42	
Guadeloupe		kg/trip	2010-2012	3.9	3.2	-18	3.2	3.9	-16	(8)
Honduras		ton/ship	1980-1995	23.0	4.3	-81	16.2	13.4	-81	(4)
			1995-2001	8.9	5.1	-43	6.0	7.3	-20	(3)
Jamaica		kg/day	1997-2007	199	74	-63	129	112	-54	(14)
Martinique		kg/trip	1987-2013	2.7	2.7	---	0.2	1.1	-82	(8)
Mexico (Quint. Roo)		ton/boat-day	1966-1995	8.4	0.9	-89	3.7	0.5	-43	(5)
Nicaragua	total	ton/boat-year	1990-2009	76.2	63.3	-17	41.3	44.7	-17	(15)
	comm.		1970-1996	57.8	34.0	-41	38.9	32.0	-41	(5)
	traps	kg/boat-day	1997-2001	111	66	-41	107	66	same	(3)
	divers			69	42	-39	69	56	-13	
Puerto Rico		kg/trip	1984-2003	12	7	-42	12	5	-44	(9)
Turks & Caicos		kg/boat-day	1966-2001	166	55	-67	166	61	-47	(3)
US Virgin Islands		kg/trip	1974-2003	75	5	-93	34	5	-50	(9)
Venezuela		kg/trap-day	1983-1988	0.12	0.06	-50	0.12	0.08	same	(5)
		kg/trap-year	1994-1997	93.2	90.8	-3	90.8	80.2	-24	
Brazil		kg/trap-day	1965-1997	0.94	0.10	-89	0.94	0.10	-89	
			1965-2004	0.94	0.08	-91	0.94	0.04	-96	(10)

Ref.

(1) Horsford et al., 2014	(6) Alzugaray and Puga, 2012
(2) Millares and López, 1974	(7) Pinteneles et al., 2016
(3) FAO, 2003	(8) FAO, 2015
(4) FAO, 2001	(9) SEDAR8, 2005a.
(5) Prada et al., 2005	(10) Silva and Fonteles-Filho, 2011

(12) Góngora, M. 2009. Assessment of the spiny lobster (*Panulirus argus*) of Belize based on fishery-dependent data. *UN Univ. Reykjavik, Iceland, Fish. Training Program, Final Project Report*, 38 *p.*

(13) Buesa, R.J., 2017. The Florida spiny lobster (*Panulirus argus*) fishery: a discussion of current status and management perspectives (Research report). *81 p.*

(14) Morris, R.A. 2010. Analysis of the Jamaican industrial spiny lobster (*Panulirus argus*) fishery. *UN Internat. Univ., Final Project Report, Reykjavik, Iceland. 44 p*

(15) Chang Bennet, R. 2010. The spiny lobster fishery in Nicaragua: a socio-economic system approach to resource management. *MS Thesis*, 81 *p.* (https://library.umaine.edu /theses/pdf/restricted/BennettRC2010.pdf)

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also allowing comparisons between these three mentioned fishing areas' spiny lobster abundances.

This common CPUE (kg/trap-day) during the 1965-1997 period allows comparing lobsters abundances in 1985 between Belize (0.76 kg/trap-day), Florida (0.052 kg/trap-day), and Brazil (0.15 kg/trap-day) concluding that during that year Belize's spiny lobster abundance was 15 times Florida's, and 5 times Brazil's. In 1997 Belize's abundance (0.54 kg/trap-day) had declined 29%, but was still 11 times Florida's (0.047 kg/trap-day), and 5 times Brazil's (0.10 kg/trap-day). The 1985 to 1997 Belize's 29% spiny lobster abundance decline coincided with a 9% effort reduction, while Brazil's 33% abundance decline coincided with a 38% effort increase both indicating Belize's populations in better condition than Brazil's, their total landings differences during the period (Brazil's 17 times Belize's) mostly consequence of Brazil's shelf being 11 times larger. These data also indicate that Florida's spiny lobster population's abundance was the lowest of the three, Brazil's being 3.4 times Florida's in 1985, and 2.1 times in 1997. By 2004 Florida's abundance was 11% less than in 1997 (0.042 kg/trap-day), and Brazil's was reduced 20% (0.08 kg/trap-day) from its 1997 value but still was twice that of Florida's despite Brazil's effort (80 million traps*days) being 35 times larger than Florida's 2.3 million, underscoring again the important role shelves' surfaces play in these fisheries with Brazil's shelf being 5.8 times larger than Florida's.

Despite any possible shortcomings, any CPUE unit consistent use allows characterizing how a lobster population's relative abundance evolved during a given time period and, except for Martinique (1987-2013) whose CPUE remained unchanged, every other fishing area (Table 5) presents decreased values, from -3% in Venezuela (1994-1997), to -93% in US Virgin Islands (1974-2003), for an overall -48% collective reduction.

Table 5 also summarizes effort variations during 31 periods/areas, with increased or same effort in only 6 at the end of the included periods, while in the rest it was less than at its peak. Only in 8 cases CPUE values were higher at peak effort (Bermuda, Northern Cuba, Guadeloupe, Honduras, Martinique, and Nicaragua), while in the rest its highest values coincided with lowest effort. Except for Belize whose effort increased 6% in 2009, and for Colombia's catches in the Luna Verde archipelago area that grew +28 %, in 1993-1997, all other most recent effort data show reductions, from -10% in Belize (1997), to -96% in Brazil (2004), for an overall -44% collective reduction, not statistically different ($P > 0.55$) to the -48% CPUE collective reduction. These CPUE reductions are reflected in a collective 38% landings decreased from a regional maximum of 52,000 mt, the unanswered question being: are landings reductions driven by abundance reductions, by effort reductions, or by their combination? The most likely answer is that abundances have steadily declined during each fishery's historical evolution increasing productions costs, causing effort reductions.

Spiny lobsters exports

Most artisanal marine fisheries essential objective is to provide food locally, but spiny lobsters' catches have evolved differently, from an almost "undesirable and worthless" by-catch in 1895 used to bait snapper traps in Florida (Crawford, and De Smidt, 1922) and elsewhere, to a cheap and not very popular source of food in some coastal communities, to a highly priced catch for local fisheries', and a hard currency source for every area where spiny lobsters are harvested. This later objective explains why even when spiny lobsters harvests kept increasing since 1950 (increasing supply) demand also increased as a hard currency source, and especially now, when supplies dwindle. In 1950 total Western Central Atlantic spiny lobster fisheries landings were 2957 mt, peaked with 43,028 mt in 1999, decrease to 31,902 mt in 2013, all the while *ex vessel* prices in Florida, usually an international prices good indicator, grew 42 fold, from 0.42 US$/kg in 1950, to 17.76 $/kg in 2015 (https://www.st.nmfs.noaa.gov/commercial-fisheries/commercial-landings/annual-landings/index).

Continuous prices increases had caused the many exploitation "excesses" these fisheries have experienced, explaining why, starting in 1957 when Turk & Caicos fishery peaked, by 2009 almost all (35) of the 38 fisheries peaked, the remaining 3 doing so in 2010 (Guadeloupe), and 2011 (St. Kitts & Nevis, and Trinidad). For the 1950-2009 period there is a positive and significant correlation (R^2 = 0.89; P < 0.005) between *ex vessel* prices (US$/kg), and the number of Caribbean peaking fisheries.

In the 1970s the average price sextupled to 3.26 US$/kg, and growing US and Europe consumers' acquisitive power determined an increased demand unsatisfied by existing catches, transforming spiny lobsters into a "luxury commodity". Prices exacerbated by the mid-2010s when Asian markets (especially Hong Kong, Japan, and the P.R. of China) started "live spiny lobsters" commercialization and soaring prices rose to close to 40 US$/kg (Figure 4).

Increasing catches exports to satisfy international markets demands, combined with the fact that spiny lobsters catches have essentially been a source of hard currency, determined most of the existing fishing excesses in most countries, except probably for the Dominican Republic where it seems a decision was made to offer *P. argus* as a "luxury staple" to its ever growing tourists numbers which increased from 1.3 Million in 1988 (Crespo and Suddaby, 2000) to 6.1 Million in 2016 (Holschuh, 2017). Selling lobster catches at restaurants eliminated international intermediaries increasing local profitability, but this approach required creating industrial fishing enterprises and extending fishing activities beyond Dominican Republic's jurisdictional waters' spiny lobster small populations, resulting in poaching activities from 2002-2014 in Bahamian territorial waters which generated legal complaints against these illegal activities.

Table 6 summarizes export rates as catches percentages for 15 large producers responsible for 97% of all *P. argus* harvests during 1950-2013 (FAO, 2014) showing that, as average, 68% of their combined catches during some periods were exported, but in all reality, that percentage is smaller for actual total harvests by several countries are larger due to spiny lobsters catches in the Pacific Ocean fishing areas besides those in the Caribbean.

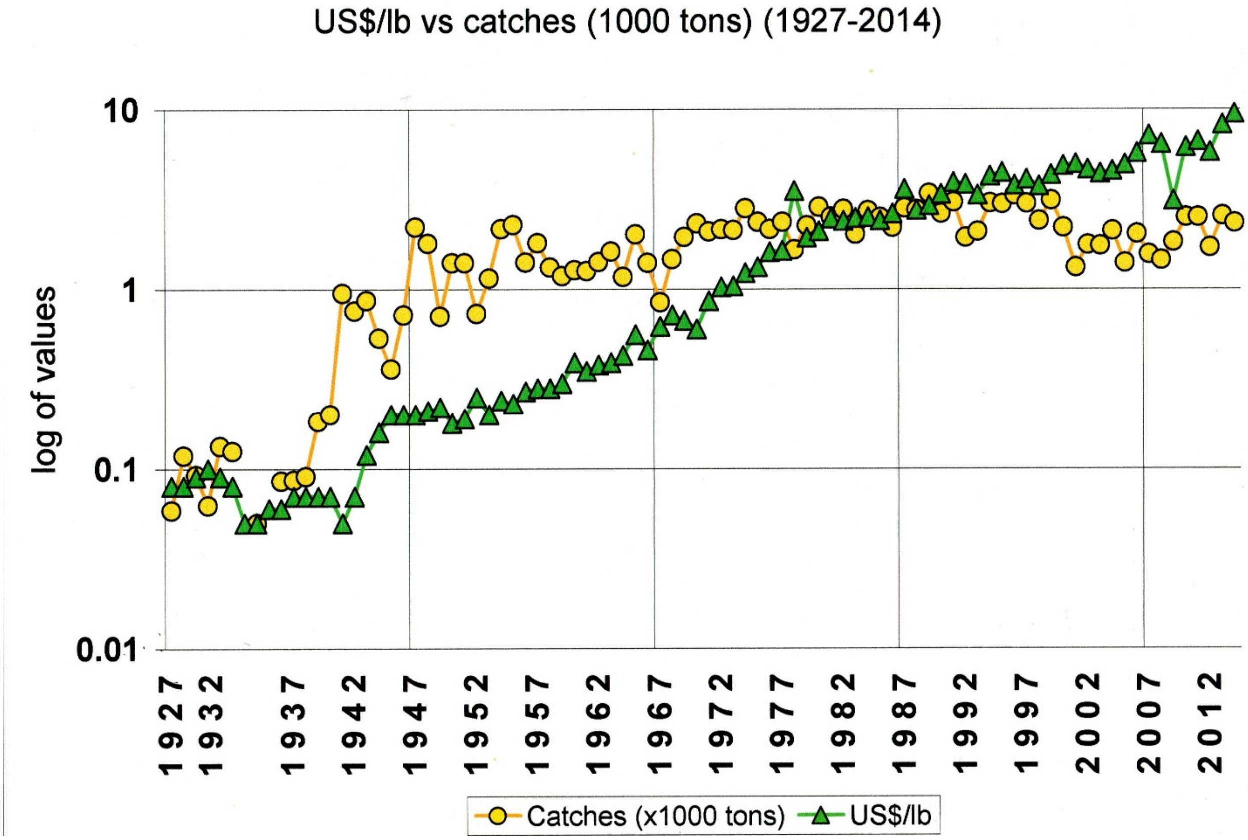

Figure 4 - Florida fishery: catches (59 to 2231 mt) vs. ex-vessel price (0.08 to 9.50 US$/lb)

For example, total US spiny lobsters catches have to include, besides *P. argus* catches in Florida, Puerto Rico, and the US Virgin Islands, those harvested in California, Hawaii, the Commonwealth of Northern Marianas, American Samoa, and Guam, all in the Pacific Ocean. To the 76,062 mt of *P. argus* from Florida, Puerto Rico, and the US Virgin Islands harvested during the 1985-2009 period in the Caribbean (Table 6), it is necessary adding 7496 mt of *Panulirus interruptus* (J.W. Randall, 1840) harvested in California. During the same period Hawaiian spiny lobsters fishery targeted two species, the endemic *P. marginatus* (Quoy & Gainard, 1825) locally known as "red ula" with 18% of catches, and *P. penilillatus* (Olivier, 1791) or "green ula", with the remaining 82% of their 1155 mt combined harvest. Spiny lobster harvests in in the Commonwealth of Northern Marianas for the same period amounted to 44 mt; 30 mt from American Samoa, and 16 mt of mostly *P. penicillatus* (Hensley and Sherwood, 1993) from Guam, for a 84,803 mt grand total of spiny lobsters fished in the United States, reducing to 23% its export rate. Similarly, during the 2014-2015 period it is necessary adding 780 mt from California, and 6 mt total from the Western Pacific, increasing the 5572 mt in the Caribbean to a total of 6358 mt, and reducing to 44% its export rate.

Six Caribbean countries (Colombia, Costa Rica, Honduras, Mexico, Nicaragua, and Panama) also harvest Pacific spiny lobsters, mostly *P. interruptus*, their combined Caribbean

Table 6 - Export volumes from 15 countries during different periods

Country	Period	Catch (mt)	Exported		Ref.
			mt	%	
Antigua & Barbuda	2011	229	46	20	(1)
Bahamas	1997	7949	2351	90	(2)
	1998-2002	31,502	27,479	87	(3)
	2013	6088	5451	90	(1)
Belize	1995	608	431	71	(2)
	1998-2001	2901	2882	99	(3)
	2013	652	548	84	(1)
Brazil	1991-2010	162,468	160,843	99	(1)
Colombia	1992-1995	2026	1606	79	(3)
Cuba	1985-2001	167,743	167,743	100	(3)
Dominican Rep.	1998-2002	5424	330	6	(3)
	2007-2013	12,375	1851	15	(1)
Haiti	2005-2013	4650	4185	90	(1)
Honduras	1996-2002	22,834	20,352	89	(3)
	2003-2013	29,687	29,687	100	(1)
Jamaica	1993	150	134	89	(2)
	1995-2001	2353	1347	57	(3)
Mexico (to USA)	1972-2009	28,512	2634	9	(4)
Nicaragua (to USA)	1972-2009	104,137	6333	6	(4)
	1996	8026	4185	52	(2)
St. Vincent & Grenadines	2004	38	34	89	(5)
Turks & Caicos	1990-2000	3395	1120	33	(3)
USA - Florida, Pto. Rico, and US Virgin Islands	1985-2009	76,062	19,340	25	(4)
	2014-2015	5272	2817	51	(6)
Total during indicated periods		685,081	463,729	68	

(1) FAO, 2015	(4) Vondruska, 2010
(2) FAO, 2001	(5) Headley and Singh-Renton, 2009
(3) FAO, 2003	(6) NOAA, 2017

harvests being 3.2 times larger than in the Pacific during 1950-2013 (FAO, 2014), although Mexico's Pacific harvests during 1985-2010 were almost twice larger as those in the Caribbean.

Regional fisheries' perspective

Every Caribbean and Brazilian spiny lobster population is essentially fully or over-exploited, its survival requiring efficient fishing effort controls, and population sizes management programs, and renewal rates calculations. Given their self-sustained condition, Brazilian populations' potential can be determined by conventional maximum sustainable yield methods, but the Caribbean interconnected meta-population's characteristic precludes this approach leaving, as only viable alternative, calculating how many of post-larvae arriving to each fishing area survive to become recruits to the area survive to become recruits to the fishery.

Table 7 - Florida post-larvae / collector (CPUE) vs. total catches with traps variations with a 2 years gap (1986-2004)

Y E A R	Post larvae (PL) (n)	Collectors (n)	PL/collector (1986-2004)	PL/collector year/period	Landings (1988-2006)	
					(2 years gap)	year/period
1 9 8 6	255	17	15	0.94	2480.3	1.01
1 9 8 7	1056	88	12	0.75	2169.0	0.89
1 9 8 8	872	76	11	0.69	2847.8	1.17
1 9 8 9	1114	82	14	0.88	2796.3	1.14
1 9 9 0	1415	104	14	0.88	3377.0	1.38
1 9 9 1	3194	376	8	0.50	2638.6	1.08
1 9 9 2	2117	229	9	0.56	1570.7	0.64
1 9 9 3	2643	256	10	0.63	1491.7	0.61
1 9 9 4	3332	238	14	0.88	2290.9	0.94
1 9 9 5	3360	259	13	0.81	2290.9	0.94
1 9 9 6	3940	207	19	1.19	3013.7	1.23
1 9 9 7	3801	196	19	1.19	3373.0	1.38
1 9 9 8	4491	167	27	1.69	3018.1	1.23
1 9 9 9	4848	198	24	1.50	2446.2	1.00
2 0 0 0	4673	217	22	1.38	3135.7	1.28
2 0 0 1	3894	192	20	1.25	2280.9	0.93
2 0 0 2	4113	201	20	1.25	1310.0	0.54
2 0 0 3	3631	213	17	1.06	1774.7	0.73
2 0 0 4	1314	73	18	1.13	2126.7	0.87
T O T A L	54063	3389	16	1.00	46432.2	
Average	2845	178	16	1.00	2443.8	

This approach is standard procedure in the Western Australia's *P. cygnus* fishery where annual postlarval abundance is used to estimate landings 3-4 years later (Caputi et al., 1995; Caputi et al., 2014). In Baja California (Mexico) *P. interruptus* postlarval abundance has been used to calculate landings 5 years later (Arteaga-Ríos et al., 2007), and Koslow et al. (2012) have found that *"Landings of the California* [spiny lobster *P. interruptus*] *fishery were significantly correlated with the abundance of stage-1 phyllosoma at lags of 7 and 8 years, but not at other lags"*.

Regarding *P. argus* an abundant literature dealing with post-larval arrivals through the Caribbean exists (Buesa, 2007; Butler et al., 2010) but harvest calculations based on abundance and survival of post-larvae, to juveniles, to recruits have been done only for the southwestern Cuban shelf (Cruz et al., 2007).

Figure 5 (from SEDAR 2005b; pp 135) shows a 50% coincidence between "puerulii" and juveniles relative abundances variations with 1year gap in between.

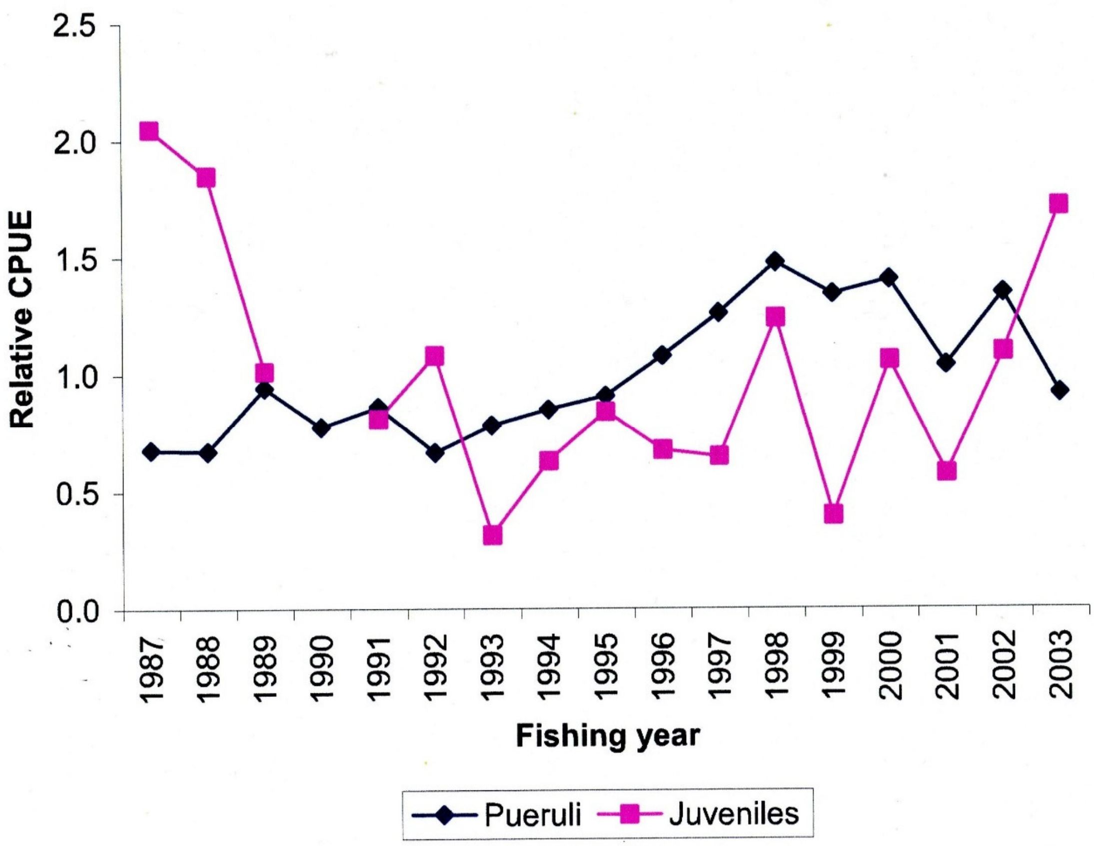

Figure 5 - [From SEDAR (2005b) described as]: "*Comparison of number of puerulii per collector and the number of juvenile lobsters from Middle Keys one year later. The year in the plot is the settlement year and the juveniles are from one year later"*.

Figure 6 shows 1986-2004 post-larvae catches from collectors situated at Big Munson Key and Long Key (Florida Keys) (SEDAR 2005b; 2010) and trap catches (1988-2006) 2 years later (Table 7) with a 61% coincidence between both annual relative abundance curves in this case. Trap catches were used because this is a passive gear totally dependent on spiny lobsters' abundance and the probability of encounter between traps and lobsters. Both relations indicate that post-larvae become juveniles after 12 months of arriving, and are recruited to the fishery 12 months later. If the 8-10 month duration of larval period (Buesa, 1972) is added, these data suggest that juvenile lobsters are approximately 20-22 months old, and recruits to the fishery are 32-34 months (almost 3 years) old, both after eclosion. It seems evident that if further research is done, it is very likely that Florida spiny lobster fishery could be managed using post-larval arrival data.

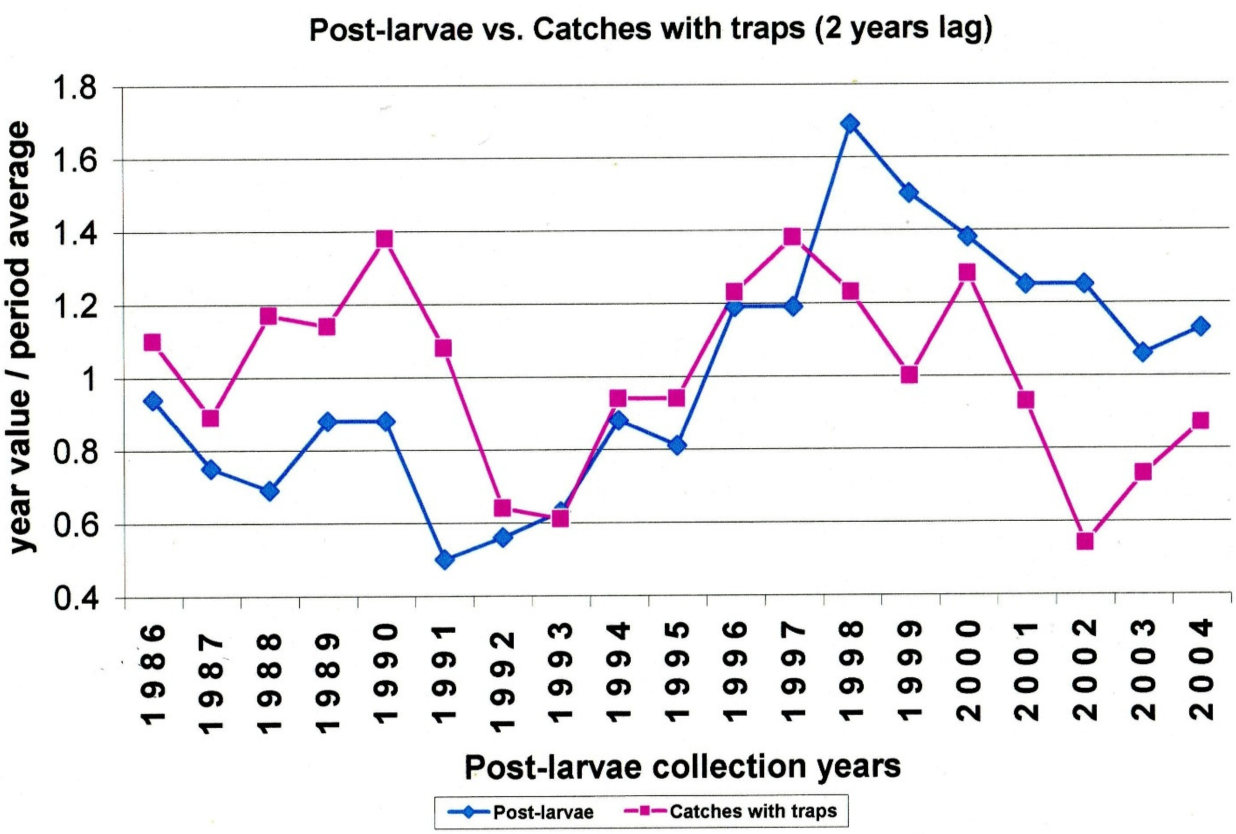

Figure 6 - Comparison of number of puerulii per collector at the settlement year, and Florida trap fishery annual catch two years later (Table 7). Both annual relative abundance curves show a 61% coincidence.

Conclusions

Fishing areas' geographical position and their total surfaces have proven to be more relevant to their total harvests amounts, than their theoretical characterization as larvae exporters or importers. Caribbean spiny lobsters populations' overexploitation present situation requires, as a collective international responsibility, that each Caribbean country contribute to the enrichment of the common larval pool to assure the regional survival, only obtainable by allowing every spiny lobster in every fishery to reach maturity and reproduce. These two fundamental objectives can only be obtained by enforcing minimum size limits, a permanent berried females capture ban, and reproduction oriented closed seasons throughout the whole Caribbean Basin. It is also imperative to acknowledge that once a post-larvae arrives at any given fishing area, irrespective of where the larval production took place, it becomes part of the national patrimony to be shared by all local fishermen and should be protected as such.

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